

HARDWARE

TASK 1:

List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

- **Motherboard**

The main circuit board that connects all components of the computer. It allows communication between the CPU, RAM, storage, and other hardware.

- **CPU (Central Processing Unit)**

Known as the brain of the computer, it processes instructions and performs calculations. It controls the overall functioning of the system.

- **RAM (Random Access Memory)**

Temporary memory used to store data that is actively being used. It helps in faster performance and multitasking.

- **GPU (Graphics Processing Unit)**

Responsible for rendering images, videos, and graphics. It is especially important for gaming and graphic-intensive tasks.

- **Storage Drive (HDD/SSD)**

Stores all data permanently, including the operating system and files. SSDs are faster, while HDDs offer more storage at lower cost.

- **Power Supply Unit (PSU)**

Supplies electrical power to all components of the computer. It converts AC power from the outlet into usable DC power.

- **Cooling System (Fans/Heat Sink)**

Prevents overheating by dissipating heat from components like the CPU. It ensures stable and efficient system performance.

- **Expansion Cards**

Additional components like sound cards or network cards added to improve functionality. They are installed in expansion slots on the motherboard.

- **Cabinet (Case)**

The outer enclosure that houses and protects all internal components. It also helps in airflow and cooling.

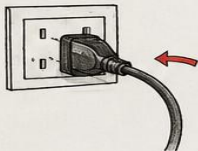
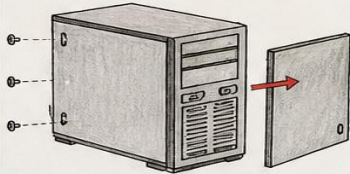
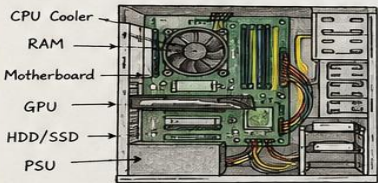
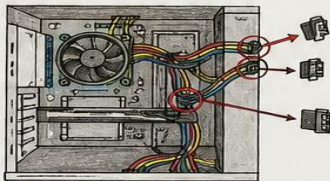
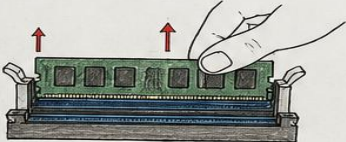
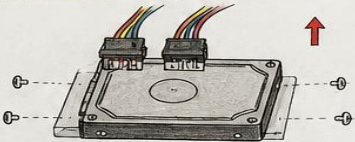
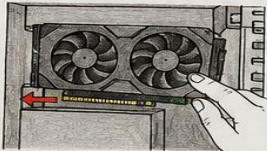
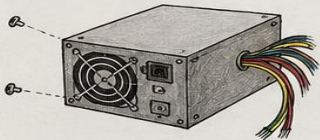
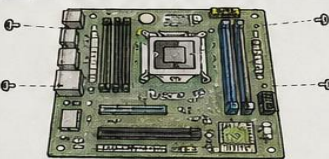
- **Cables and Connectors**

Used to connect different components and supply power or data signals. Proper cable management helps in better airflow and performance.

TASK 2:

Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

DISASSEMBLING A DESKTOP PC – STEP BY STEP	
<u>Step 1: Unplugging the Power Supply</u>  Switched off the computer and unplugged the power cable from the electrical outlet. This ensures safety before starting.	<u>Step 2: Removing the Side Panel</u>  Used a screwdriver to remove the screws at the back of the cabinet and carefully slid off the side panel.
<u>Step 3: Observing Internal Components</u>  After opening the case, I examined the layout and identified all major parts.	<u>Step 4: Disconnecting Power Cables</u>  Carefully unplugged power supply cables connected to the motherboard, GPU, HDD/SSD and other components.
<u>Step 5: Removing RAM (Memory)</u>  Pressed the clips on both sides of the RAM slot outward. The RAM stick pops up, then I removed it carefully.	<u>Step 6: Removing Storage Drive (HDD/SSD)</u>  Disconnected the SATA and power cables. Then unscrewed the drive from the bay and took it out.
<u>Step 7: Removing GPU (Graphics Card)</u>  Unscrewed the GPU from the case, released the PCIe slot lock and carefully pulled the graphics card out.	<u>Step 8: Removing Cooling Fan and Heat Sink</u>  Unplugged the fan cable from the motherboard. Unscrewed the heat sink and gently removed it from the CPU.
<u>Step 9: Removing the Power Supply Unit (PSU)</u>  Disconnected all remaining cables from the PSU. Then unscrewed it from the case and slid it out.	<u>Step 10: Removing the Motherboard</u>  After disconnecting all cables and components, I unscrewed the motherboard and lifted it out carefully.
<u>Conclusion:</u> This step-by-step process helped me understand how different components of a desktop PC are connected and how to safely disassemble them. Proper handling and careful removal are essential to avoid damage and ensure safe reassembly.	<u>Safety Tips:</u> • Always unplug the power cable first. • Handle components with care. • Keep screws in a safe place. • Avoid touching circuits directly.

Step 1: Unplugging the Power Supply

Caption: Switched off the computer and unplugged the power cable from the electrical outlet.

Before starting, I ensured the PC was completely powered off and disconnected from electricity to avoid any risk of electric shock or damage to components.

Step 2: Removing the Side Panel

Caption: Opened the cabinet by removing screws and sliding the side panel. Using a screwdriver, I removed the screws at the back of the cabinet and carefully slid off the side panel to access the internal components.

Step 3: Observing Internal Components

Caption: Identified major components like motherboard, CPU, RAM, storage, and PSU.

After opening the case, I examined the layout and identified all major parts to understand their placement before removing anything.

Step 4: Disconnecting Power Cables

Caption: Disconnected power cables from motherboard, storage devices, and GPU.

I carefully unplugged all power supply cables connected to components such as the motherboard, hard drive, and graphics card.

Step 5: Removing RAM (Memory)

Caption: Removed RAM by unlocking clips on both sides.

I gently pressed the clips on the RAM slots outward, allowing the RAM sticks to pop up, and then removed them carefully.

Step 6: Removing Storage Drive (HDD/SSD)

Caption: Unscrewed and disconnected the storage device.

I disconnected the SATA and power cables, unscrewed the drive from its bay, and carefully took it out.

Step 7: Removing GPU (if present)

Caption: Removed graphics card from PCIe slot.

After unscrewing the GPU from the case, I released the PCIe slot lock and carefully pulled the graphics card out.

Step 8: Removing Cooling Fan and Heat Sink

Caption: Detached CPU cooling system.

I unplugged the fan cable and unscrewed the heat sink, then gently removed it from the CPU.

Step 9: Removing the Power Supply Unit (PSU)

Caption: Unscrewed and removed the PSU from the cabinet.

I disconnected all remaining cables and unscrewed the PSU from the case before sliding it out.

Step 10: Removing the Motherboard

Caption: Carefully unscrewed and removed the motherboard.

After disconnecting all cables and components, I unscrewed the motherboard and lifted it out carefully to complete the disassembly.

TASK 3:

Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

Step 1: Installing the Motherboard

Explanation: I carefully placed the motherboard inside the cabinet and aligned it with the screw holes, then secured it using screws.

Why order matters: The motherboard is the main base, so all other components are connected to it; it must be installed first.

Step 2: Installing the Power Supply Unit (PSU)

Explanation: I inserted the PSU into its slot and fixed it with screws, ensuring proper positioning for cable management.

Why order matters: Installing the PSU early helps in routing cables easily to other components later.

Step 3: Installing the CPU

Explanation: I placed the CPU into the socket carefully, aligning it properly, and locked it in place.

Why order matters: The CPU must be installed before the cooler and RAM, as it is central to system operation.

Step 4: Attaching the Cooling System (Heat Sink & Fan)

Explanation: I applied thermal paste and mounted the heat sink and fan over the CPU, then connected the fan cable.

Why order matters: Cooling is essential immediately after CPU installation to prevent overheating.

Step 5: Installing RAM (Memory)

Explanation: I inserted the RAM sticks into the slots and pressed until the clips locked them securely.

Why order matters: RAM should be installed before adding larger components like GPU to ensure easy access.

Step 6: Installing Storage Devices (HDD/SSD)

Explanation: I mounted the storage drive in its bay and connected SATA and power cables.

Why order matters: Storage installation is easier before the case becomes crowded with cables and cards.

Step 7: Installing GPU (Graphics Card)

Explanation: I inserted the GPU into the PCIe slot, secured it with screws, and connected power cables if required.

Why order matters: The GPU is large and can block access, so it is installed after smaller components.

Step 8: Connecting All Power and Data Cables

Explanation: I connected cables from the PSU to the motherboard, GPU, and storage devices, along with front panel connectors.

Why order matters: Proper connections ensure all components receive power and communicate correctly.

Step 9: Cable Management

Explanation: I arranged and tied cables neatly to improve airflow and avoid obstruction.

Why order matters: Good cable management prevents overheating and makes future maintenance easier.

Step 10: Closing the Cabinet and Testing

Explanation: I reattached the side panel, plugged in the power cable, and turned on the PC to check functionality.

Why order matters: Final testing ensures everything is properly installed before regular use.

TASK 4:

Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

A. Safety Precautions

1. **Switch Off and Unplug Power Supply**
Always turn off the PC and unplug it from the power source to prevent electric shock and damage to components.
2. **Use an Anti-Static Wrist Strap**
Prevents electrostatic discharge (ESD), which can damage sensitive components like the motherboard and RAM.
3. **Work on a Clean and Dry Surface**
Avoid dust, liquids, or clutter to reduce the risk of short circuits or losing small parts.
4. **Handle Components by Edges**
Touching circuits or pins can damage components; holding edges keeps them safe from oils and static.
5. **Avoid Working on Carpeted Floors**
Carpets generate static electricity, increasing the risk of damaging internal parts.
6. **Keep Screws and Small Parts Organized**
Store screws in a container to avoid losing them, ensuring smooth reassembly.
7. **Do Not Force Components**
Apply gentle pressure when installing parts to avoid breaking connectors or slots.

B. Tools Required

Screwdriver (Phillips Head)

Used to open the cabinet and remove or install components securely.

Anti-Static Wrist Strap or Mat

Protects components from static electricity during handling.

Small Container or Magnetic Tray

Keeps screws and small parts safe and organized during the process.

Thermal Paste

Applied between CPU and heat sink to improve heat transfer and prevent overheating.

Cleaning Cloth or Compressed Air

Used to remove dust from components, ensuring better performance and cooling.

Cable Ties (Zip Ties)

Helps in organizing cables for better airflow and neat arrangement.

Flashlight (Optional)

Useful for seeing small connectors and parts inside the cabinet clearly.

Conclusion

Following proper safety precautions and using the right tools ensures a safe and efficient PC assembly or disassembly process. It helps prevent damage, improves performance, and makes the work easier and more organized.